

Table 1. Operational gene sets defining stage-specific states of MuSCs during regeneration.

Regenerative Stage	Functional Definition	Representative Markers	Key References
Quiescence	Stem cell maintenance, niche anchorage, stress resistance	Pax7, Notch3, Calcr, Foxo3, Spry1	[38–41]
Early activation	Immediate-early transcription, stress adaptation, early myogenic priming	Fos, Jun, Egr1, Atf3, Myf5, Myod1	[10,11,32,42–46]
Proliferation	Cyclin induction, S-phase entry, mitotic progression	Mki67, Ccnd1, Ccnb1, Pcna	[9,10,44,47,48]
Differentiation	Terminal myogenic commitment and contractile program	Myog, Myh3, Myh8, Mef2c	[32,51]

The complete gene lists and literature mapping are provided in Supplementary Table S1.